



**Politecnico
di Torino**

Tire Dynamics Modeling and Analysis

Bachelor's Degree in Automotive Engineering

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Abstract

This project investigates the main parameters influencing tire behavior. The first step involves evaluating the stiffness associated with longitudinal forces, lateral forces, and the self-aligning moment. These stiffness values are then used for the linear analysis of vehicle dynamics, and are obtained by processing the Magic Formula of Pacejka's model. The second step focuses on analyzing the non-linear behavior. Tire characteristics are plotted over the full range of slip and slip angle and subsequently discussed. The third step addresses the combined effect of longitudinal and lateral forces.

Contents

1	Introduction	4
2	Linear model	4
2.1	Maximum longitudinal force coefficient and slope of $F_x - \sigma$ curve	4
2.1.1	Maximum longitudinal force coefficient μ_{xp}	5
2.1.2	Slope BCDx of the curve $F_x - \sigma$ at the origin when $\sigma = 0$	6
2.2	Maximum lateral force coefficient and slope of $F_y - \alpha$ curve	7
2.2.1	Side slip stiffness BCDy with $\gamma = 0$	8
2.2.2	Maximum lateral force coefficient	9
2.2.3	Effect of the camber angle γ on the side slip stiffness BCDy	10
2.3	Self-alignment torque stiffness	10
2.3.1	Self-alignment torque stiffness BCDz with $\gamma = 0$	12
2.3.2	Effect of the camber angle γ on the self-alignment torque stiffness BCD_z	12
3	Tire non-linear model	13
3.1	Longitudinal force coefficient μ_x as a function of the slip ratio σ	13
3.2	Lateral force coefficient μ_y as a function of the slip angle α	14
3.3	Lever arm t of the lateral force F_y as a function of slip angle α	14
3.4	Gough diagram	15
3.5	Tire stiffnesses vs vertical load	16
4	Interaction between longitudinal and lateral tire-road forces	17
4.1	Lateral force F_y as a function of the longitudinal force F_x	18
4.2	Lateral force coefficient μ_y as a function of the longitudinal force coefficient μ_x	19
4.3	Side-slip stiffness C as a function of the longitudinal force F_x	19
5	Conclusions	20

1 Introduction

In this report, the formulas employed and the resulting graphs are presented and discussed as simply as possible. The reference tire considered in the study is a passenger car tire of size 205/55 R16. In the first section, the F_x - σ diagram is used to determine different values of slip stiffness and the maximum longitudinal force coefficient μ_{xp} . The side slip stiffness is evaluated from the F_y - α curves, where the maximum values of the lateral force coefficient μ_{yp} are also obtained. In addition, the influence of the camber angle on side slip stiffness is analyzed by considering values of γ ranging from $\pm 20^\circ$. The same considerations are applied to the self-aligning torque M_z . The second section addresses the non-linear behavior of the tire under different operating conditions. Vertical loads ranging from 2 kN to 10 kN are applied, and the resulting graphs of μ_y , μ_x , and t , the lever arm of the lateral force generating the self-aligning torque, are presented. The lateral behavior is then summarized in the so-called *Gough Diagram*. The last section of the project focuses on the interaction of road-tire forces in the longitudinal and lateral directions, for which polar diagrams are obtained.

2 Linear model

2.1 Maximum longitudinal force coefficient and slope of $F_x - \sigma$ curve

To compute the longitudinal force of the tire, the linear model based on Pacejka's Magic Formula was used. This model provides the longitudinal force F_x as a function of the slip σ :

$$F_x = D \sin\{C \arctan(B(1 - E)(\sigma + S_h)) + E \arctan[B(\sigma + S_h)]\} + S_v \quad (1)$$

with the coefficients:

$$\begin{cases} C = b_0 \\ D = \mu_{xp} F_z = (b_1 F_z + b_2) F_z \\ BCD_x = (b_3 F_z^2 + b_4 F_z) e^{-b_5 F_z} \\ E = b_6 F_z^2 + b_7 F_z + b_8 \\ S_h = b_9 F_z + b_{10} \\ S_v = 0 \end{cases} \quad (2)$$

where the coefficient BCD_x is the slope ($\partial F_x / \partial \sigma$) of the curve F_x - σ at the origin ($\sigma = 0$).

The coefficients and the output of the magic formula use different units: the vertical force F_z needs to be given in kN, the slip ratio σ in percentage whereas the longitudinal force F_x is given in N. The coefficients b_i used are listed in the table below.

b_0	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8	b_9	b_{10}
1.65	0	1688	0	299	0	0	0	-10	0	0

Table 1: b_i tire coefficients values

By using the previous equations, it is possible to evaluate and plot as a function of F_z :

- the maximum longitudinal force coefficient μ_{xp} ,
- the slope BCD_x of the curve F_x - σ at the origin when $\sigma = 0$.

The results are plotted for five vertical load values, specifically $F_z = 2, 4, 6, 8, 10$ kN. First, the equations are used to plot the F_x - σ characteristic of the tire:

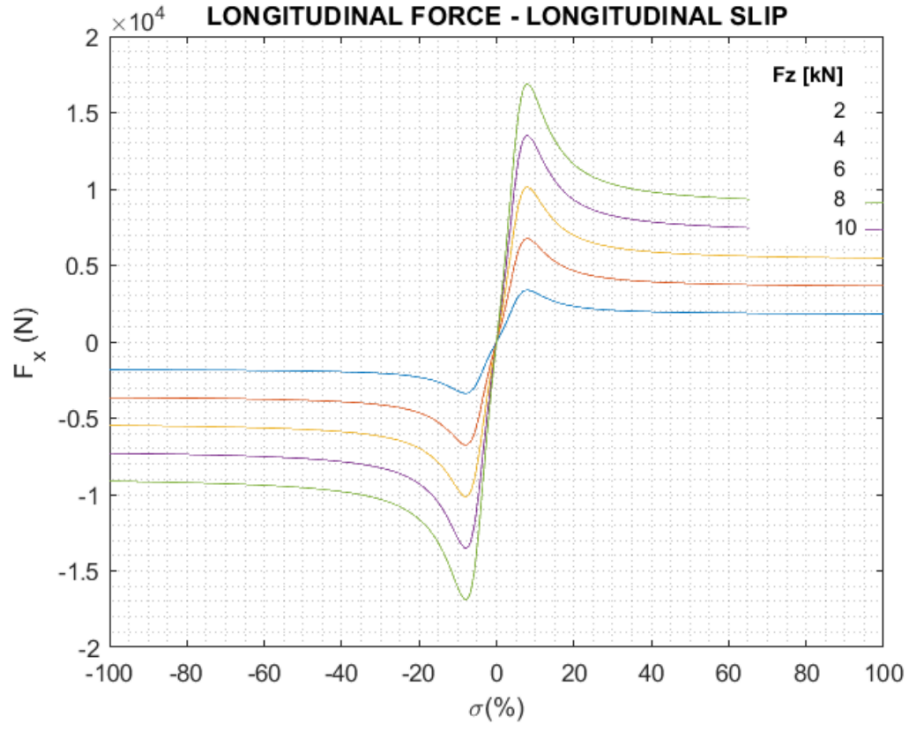


Figure 1: Longitudinal force vs tire longitudinal slip

As expected, the longitudinal force F_x increases with increasing vertical load F_z . The peak value of F_x is reached for very small values of slip σ (around 4-5%). The different curves represent the force at various vertical loads (denoted by F_z in kiloNewtons [kN]), ranging from 2 kN to 10 kN. As the vertical load increases, the peak longitudinal force also increases. This indicates that heavier loads can generate higher traction forces, up to a certain limit.

2.1.1 Maximum longitudinal force coefficient μ_{xp}

The peak longitudinal force coefficient can be evaluated starting from Equation 2:

$$\mu_{xp} = \frac{(b_1 F_z + b_2) F_z}{F_z \cdot 10^3} = \frac{(b_1 F_z + b_2)}{10^3} \quad (3)$$

The computed values of the maximum longitudinal force coefficient are listed in the table below:

F_z [kN]	μ_{xp}
2	1.688
4	1.688
6	1.688
8	1.688
10	1.688

Table 2: Maximum tire longitudinal coefficients

Plotting the results, the following graph is obtained:

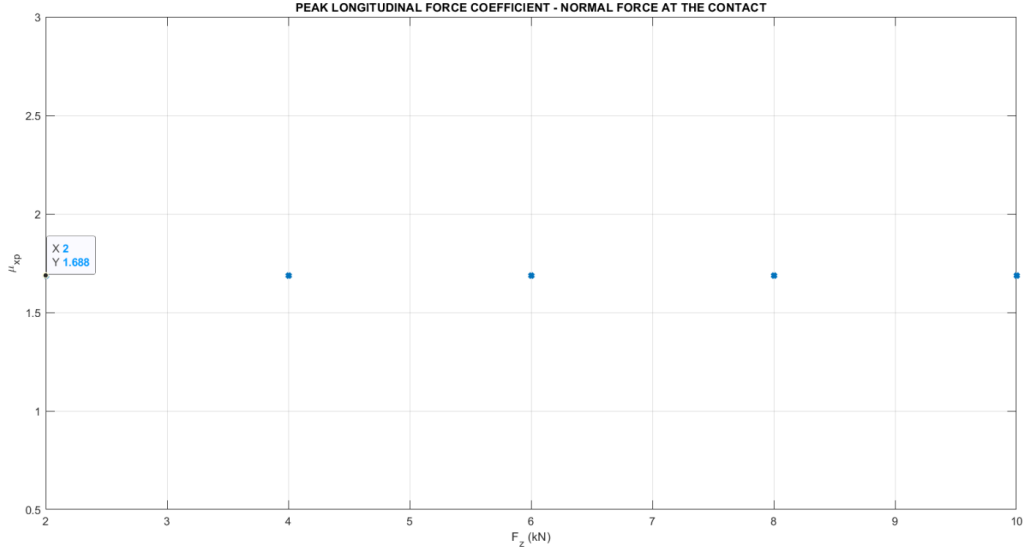


Figure 2: Peak longitudinal force coefficient vs vertical load

The tire does not exhibit a significant decrease in the peak longitudinal force coefficient with increasing vertical load, which suggests a relatively load-insensitive behavior within this range. Moreover, the results show that the peak longitudinal force is independent of the normal force, since $b_1 = 0$.

2.1.2 Slope BCD_x of the curve F_x - σ at the origin when $\sigma = 0$

By using Equation 2, the longitudinal stiffness of the tire can also be evaluated. The corresponding values are reported in the table below.

F_z [kN]	BCD_x [N/deg]
2	458
4	916
6	1374
8	1832
10	2290

Plotting the results, the following graph is obtained:

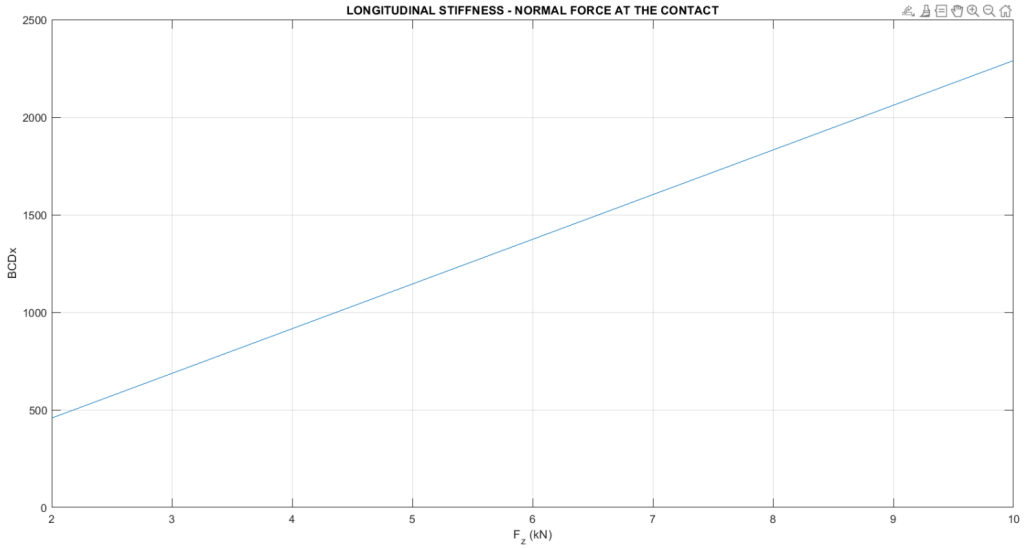


Figure 3: Longitudinal stiffness vs vertical load

This demonstrates that the slope of the F_x - σ curve at the origin depends on the vertical load F_z . The trend observed is linearly dependent on F_z .

2.2 Maximum lateral force coefficient and slope of F_y - α curve

To compute the lateral force of the tire, the linear model based on Pacejka's Magic Formula is used. This model provides the lateral force F_y as a function of the sideslip angle α and the camber angle γ :

$$F_y = D \sin\{C \arctan(B(1 - E)(\sigma + S_h)) + E \arctan[B(\sigma + S_h)]\} + S_v \quad (4)$$

with the coefficients:

$$\begin{cases} C = a_0 \\ D = \mu_{yp} F_z = (a_1 F_z + a_2) F_z \\ BCD_y = a_3 \sin\left(2 \arctan\left(\frac{F_z}{a_4}\right)\right) (1 - a_5 |\gamma|) \\ E = a_6 F_z + a_7 \\ S_h = a_8 \gamma + a_9 F_z + a_{10} \\ S_v = a_{11} \gamma F_z + a_{12} F_z + a_{13} \end{cases} \quad (5)$$

where the coefficient a_{11} is given by the relationship:

$$a_{11} = a_{111} F_z + a_{112} \quad (6)$$

The coefficient BCD_y is the slope $(\partial F_y / \partial \alpha)$ of the curve F_y - α at the origin ($\alpha = 0$) and depends on the camber angle γ .

As for the longitudinal force, the units of the coefficients and of the output of the magic formula for the lateral force are not consistent. The lateral force F_y is given in N, the vertical force F_z in kN whereas the angles α and γ are in degrees. The coefficients a_i are listed in the table below.

a_0	a_1	a_2	a_3	a_4	a_5	a_6	a_7	a_8	a_9	a_{10}	a_{11}	a_{12}	a_{13}
1.80	0	1688	4140	6.026	0	-0.3589	1	0	0	0	0	0	0

Table 3: a_i tire coefficients values

Using the previous equations, it is possible to evaluate and plot the following for five vertical load values $F_z = 2, 4, 6, 8, 10$ kN:

- the side slip stiffness BCD_y with $\gamma = 0$,
- the maximum lateral force coefficient μ_{yp} ,
- the effect of the camber angle γ on the side slip stiffness BCD_y , varying γ between $\pm 20^\circ$ and considering a single vertical load $F_z = 5$ kN.

First, the F_y - α characteristic of the tire can be plotted:

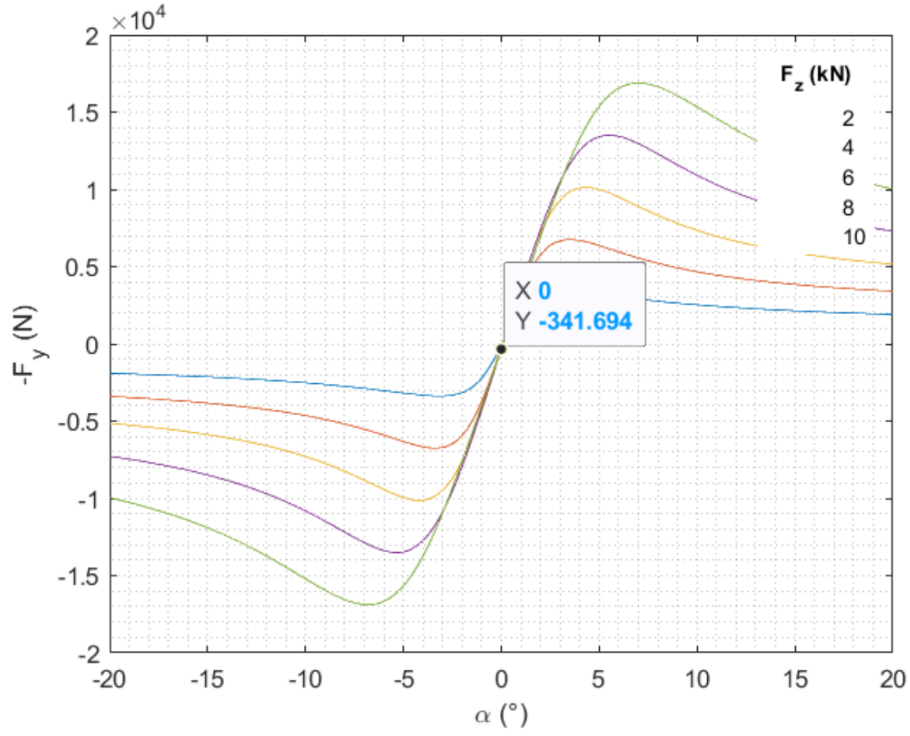


Figure 4: Lateral force vs side slip angle

As expected, the lateral force F_y increases with increasing vertical load F_z . This graph was obtained with a zero camber angle ($\gamma = 0^\circ$). At zero sideslip angle ($\alpha = 0$), the cornering force is -341.69 N. This is because the sideslip angle is defined from geometrical considerations. The results are computed assuming that positive values of the sideslip angle correspond to negative values of the lateral force.

2.2.1 Side slip stiffness BCDy with $\gamma = 0$

By using Equation 5 and setting the camber angle $\gamma = 0$, the lateral stiffness of the tire can be evaluated. The computed values are listed in the table below:

F_z [kN]	BCD_y [N/deg]
2	2475
4	3815
6	4140
8	3979
10	3660

Plotting the results, the following graph is obtained:

SIDE-SLIP STIFFNESS VS VERTICAL LOAD ($\gamma=0$)

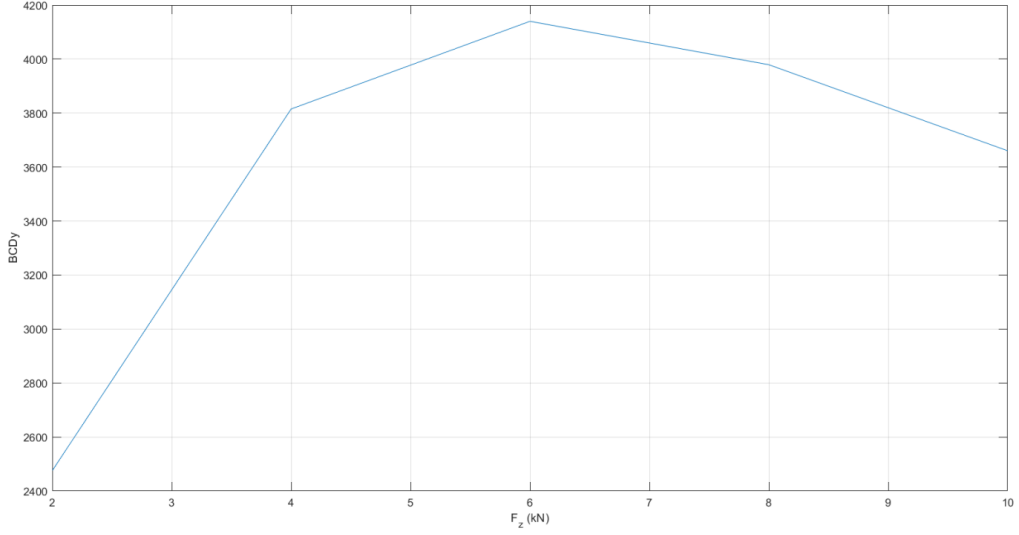


Figure 5: Side-slip stiffness vs vertical load

As expected, the side-slip stiffness peaks at 6 kN of downward force, indicating tire saturation, and then a gradual decrease after the maximum is reached. Since only very few data points were inserted, the graph seems very jagged. In the real world with more data points a continuous and smoother curve with similar behavior would be observed.

2.2.2 Maximum lateral force coefficient

The peak lateral force coefficient can be evaluated from Equation 5 as follows:

$$\mu_{yp} = \frac{(a_1 F_z + a_2) F_z}{F_z \cdot 10^3} = \frac{(a_1 F_z + a_2)}{10^3} \quad (7)$$

The computed values of the maximum lateral force coefficient are listed in the table below:

F_z [kN]	μ_{yp}
2	1.688
4	1.688
6	1.688
8	1.688
10	1.688

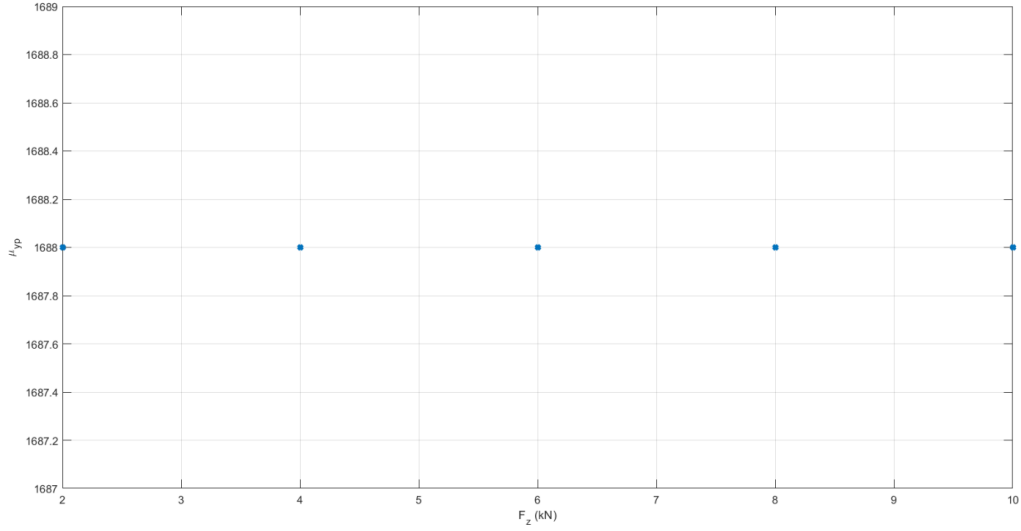


Figure 6: Peak lateral force coefficient vs vertical load

The peak lateral force does not depend on the vertical load, since the coefficient a_1 is zero, and it has a constant value of 1.68.

2.2.3 Effect of the camber angle γ on the side slip stiffness BCDy

Using Equation 5 and varying the camber angle γ between $\pm 20^\circ$, the lateral stiffness of the tire can be evaluated as a function of the camber angle, considering a single vertical load $F_z = 5$ kN.

Plotting the results the following graph is obtained:

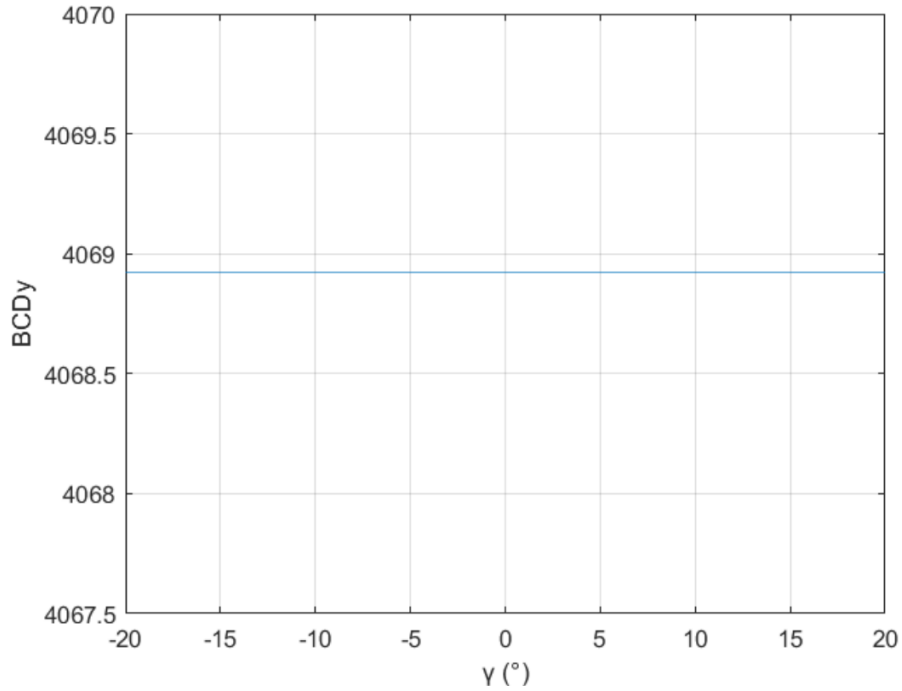


Figure 7: Side-slip stiffness vs camber angle

This reported chart shows that the side slip stiffness is independent of the camber angle, since the coefficient a_5 is zero.

2.3 Self-alignment torque stiffness

The Pacejka's magic formula gives the self-alignment torque as:

$$M_z = D \sin\{C \arctan[B(1 - E)(\sigma + S_h)] + E \arctan[B(\sigma + S_h)]\} + S_v \quad (8)$$

with the coefficients:

$$\begin{cases} C = c_0 \\ D = (c_1 F_z + c_2) F_z \\ BCD_z = -(c_3 F_z^2 + c_4 F_z) (1 - c_6 |\gamma|) e^{-c_5 F_z} \\ E = (c_7 F_z^2 + c_8 F_z + c_9) (1 - c_{10} \gamma) \\ S_h = c_{11} \gamma + c_{12} F_z + c_{13} \\ S_v = (c_{14} F_z^2 + c_{15} F_z) \gamma + c_{16} F_z + c_{17} \end{cases} \quad (9)$$

The coefficient BCD_z is the slope ($\partial M_z / \partial \alpha$) of the curve M_z - α at the origin ($\alpha = 0$) and depends on the camber angle γ .

As before, the vertical force F_z is given in kN, the side slip angle α and the camber γ are in degrees whereas the self-aligning torque M_z is in Nm. The coefficients c_i are listed in the tables below:

C ₀	C ₁	C ₂	C ₃	C ₄	C ₅	C ₆	C ₇	C ₈	C ₉
2.068	-6.49	-21.85	0.4160	-21.31	0.0294	0	-1.197	5.228	-14.84

C ₁₀	C ₁₁	C ₁₂	C ₁₃	C ₁₄	C ₁₅	C ₁₆	C ₁₇
0	0	-0.03736	0.038910	0	0	0.639	1.693

Using the previous equations, it is possible to evaluate and plot:

- the stiffness BCD_z of the self-aligning torque for five vertical loads $F_z = 2, 4, 6, 8, 10$ kN, with camber angle $\gamma = 0^\circ$
- the effect of the camber angle γ on the stiffness BCD_z , using a single vertical load $F_z = 5$ kN and varying γ within the range $\pm 20^\circ$.

First, the M_z - α characteristic of the tire can be plotted:

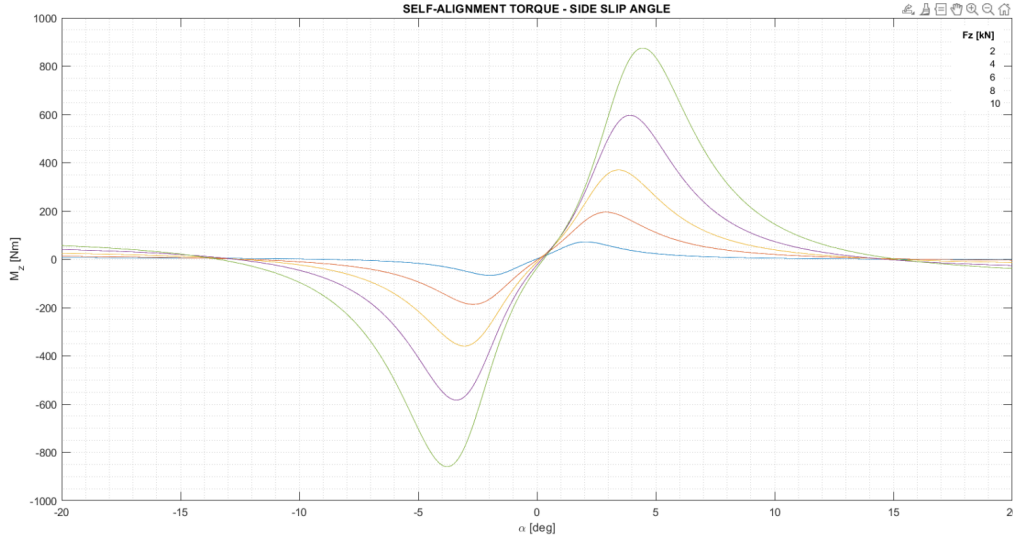


Figure 8: Self-aligning moment vs side-slip angle

The curves typically start at or near zero torque when the slip angle is zero, indicating that no lateral force is causing the tire to deflect. As the slip angle increases in either direction, negative or positive, the self-aligning torque initially increases. The torque acts to bring the tire back toward the zero slip angle position, which explains the term "self-aligning". Each curve reaches a peak, representing the maximum self-aligning torque the tire can generate. This peak corresponds to the point of maximum effectiveness of the tire in correcting its alignment with the direction of travel. Beyond the peak, the self-aligning torque decreases with further increases in slip angle, indicating that the tire is losing its self-aligning effectiveness and approaching lateral sliding, i.e. loss of grip. The position of the peak and its magnitude can vary with vertical load. Higher loads may shift the peak torque to lower slip angles, and the magnitude of the peak torque generally increases with vertical load.

2.3.1 Self-alignment torque stiffness BCD_z with $\gamma = 0$

Using Equation 9 and setting the camber angle $\gamma = 0^\circ$, the self-aligning torque stiffness of the tire can be evaluated. The computed values are reported in the table below:

F_z [kN]	BCD_z [N/deg]
2	38.61
4	69.85
6	94.62
8	113.7
10	127.8

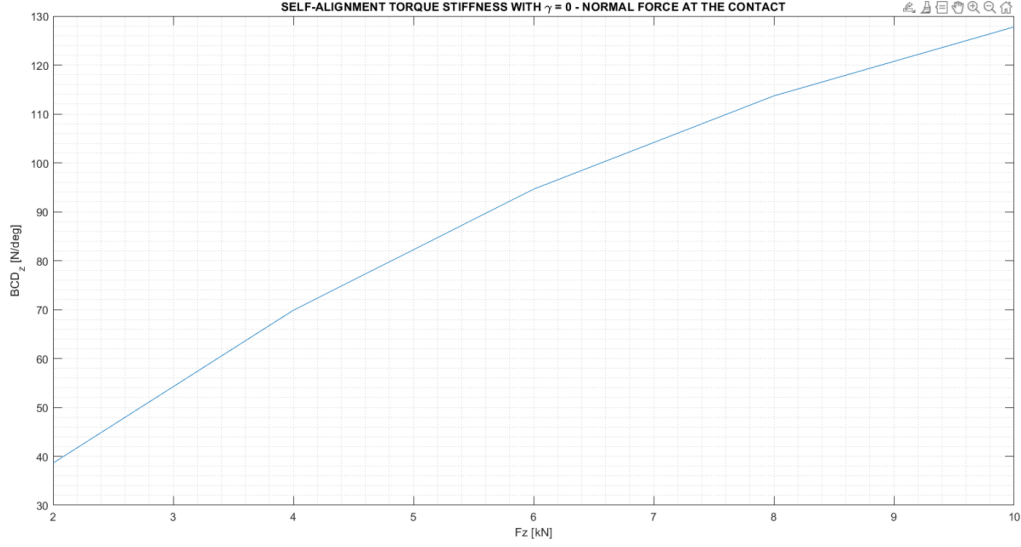


Figure 9: Self-alignment moment stiffness vs vertical load

The value of the self-alignment moment stiffness increases with the vertical load F_z . This explains why the sideslip angle at which M_z changes sign also increases with increasing F_z .

2.3.2 Effect of the camber angle γ on the self-alignment torque stiffness BCD_z

Using Equation 9 and varying the camber angle γ between $\pm 20^\circ$, the self-aligning torque stiffness of the tire can be evaluated as a function of the camber angle, considering a single vertical load $F_z = 5$ kN.

Plotting the results the following graph is obtained:

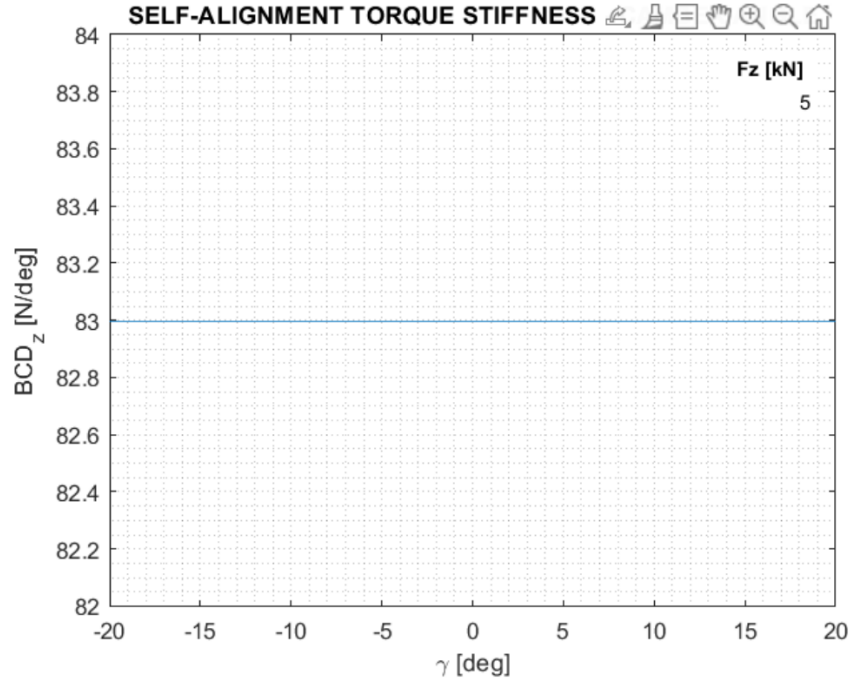


Figure 10: Self-alignment moment stiffness vs camber angle

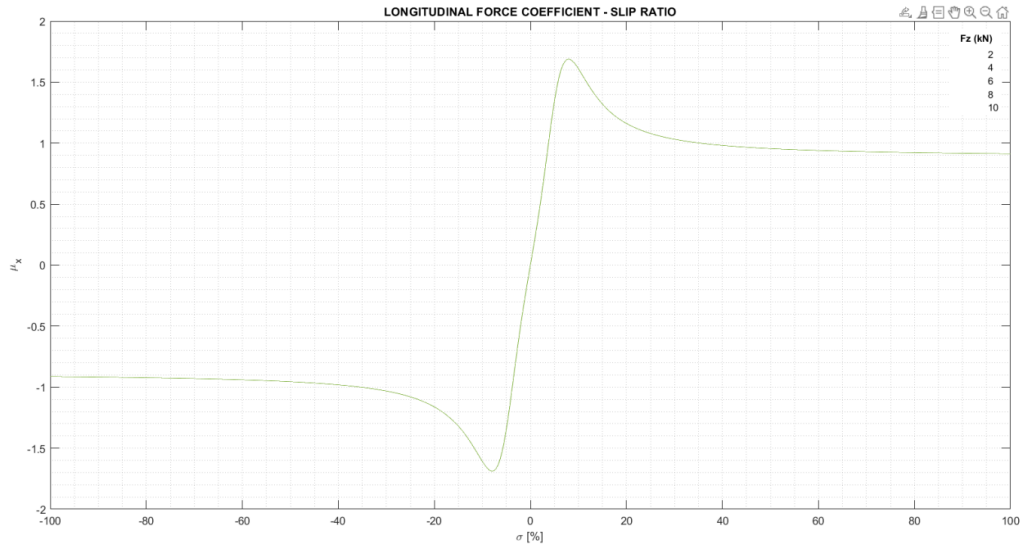
The stiffness BCD_z does not depend on the camber angle, since the coefficient c_6 is zero.

3 Tire non-linear model

The objective of this section is to estimate the non-linear behavior of the tire under different operating conditions. In the previous section, the generalized forces developed by the tire for various vertical loads over a wide range of operating conditions were computed. Based on these data, it is possible to:

- evaluate and plot the longitudinal force coefficient μ_x as a function of the slip ratio σ for vertical loads $F_z = 2, 4, 6, 8, 10$ kN
- evaluate and plot the lateral force coefficient μ_y as a function of the slip angle α for vertical loads $F_z = 2, 4, 6, 8, 10$ kN, with camber angle $\gamma = 0^\circ$
- evaluate and plot the lever arm t of the lateral force F_y , which generates the self-aligning torque M_z ($M_z = F_y t$), as a function of slip angle α , using the same vertical loads $F_z = 2, 4, 6, 8, 10$ kN with $\gamma = 0^\circ$
- plot the Gough diagram
- compute the slope of the F_x , F_y , and M_z curves using the increment method to determine their derivatives and compare these slopes with the stiffnesses BCD computed in the first part of the project.

3.1 Longitudinal force coefficient μ_x as a function of the slip ratio σ



captionLongitudinal force coefficient vs longitudinal slip

As expected, the peak longitudinal friction coefficient is confirmed to be independent of the vertical load.

3.2 Lateral force coefficient μ_y as a function of the slip angle α

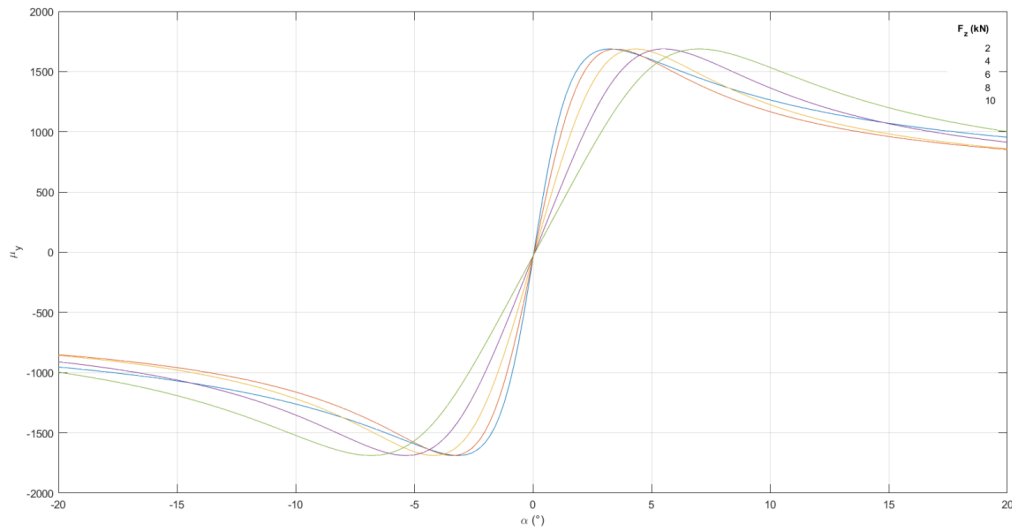


Figure 11: Lateral force coefficient vs side-slip angle

This further confirms the concept discussed in Section 2.2.2, where the lateral friction coefficient decreases slightly with increasing vertical load after reaching the saturation point. The curves do not pass through the origin, as explained in Section 2.2.

3.3 Lever arm t of the lateral force F_y as a function of slip angle α

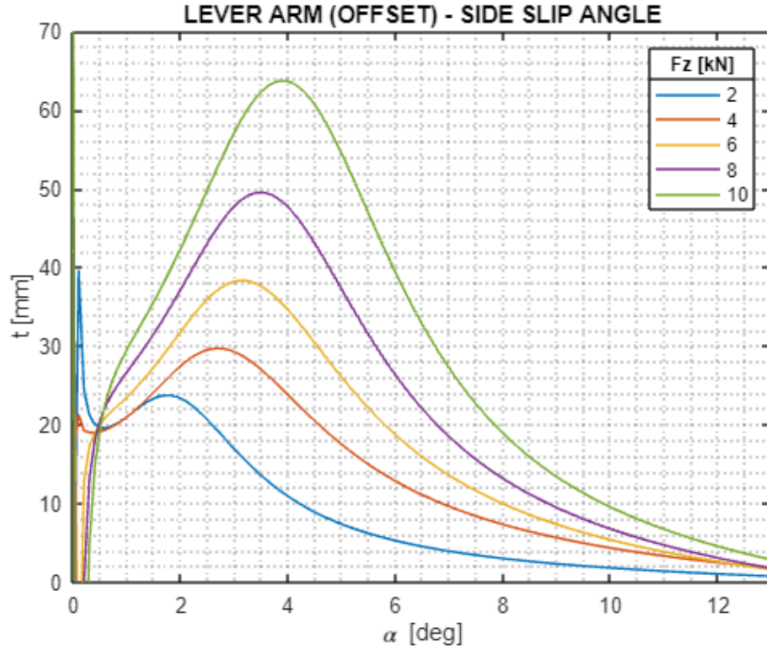


Figure 12: Lever arm vs side-slip angle

As the sideslip angle increases from zero, the lever arm also increases. This indicates that the cornering force initially acts further behind the center of the contact patch, enhancing the self-aligning torque. Each curve exhibits a peak value of the lever arm, after which it decreases as the sideslip angle continues to increase. The peak corresponds to the maximum pneumatic trail for that load, before the lateral force moves closer to the center of the contact patch or even ahead of it, thereby reducing the self-aligning torque. The peak typically occurs at a small slip angle, indicating that the maximum self-aligning effect is achieved relatively early in the slip angle range. The magnitude of the lever arm is influenced by the vertical load, with higher loads generally producing a larger peak. The exact relationship, however, may vary depending on the tire design.

3.4 Gough diagram

All results of the tire's lateral behavior can be summarized in the Gough diagram, where the lateral force F_y is plotted against the elf-aligning torque, with variations in the parameters F_z , α , and t .

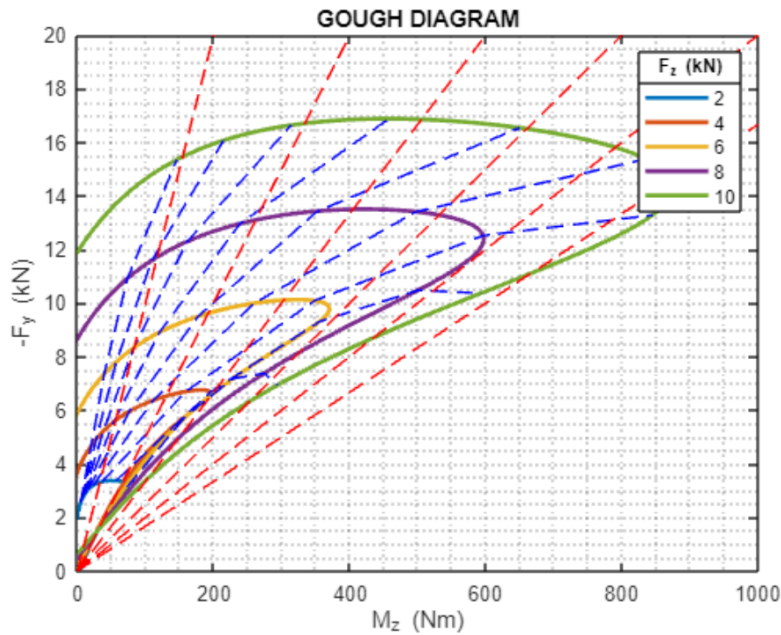


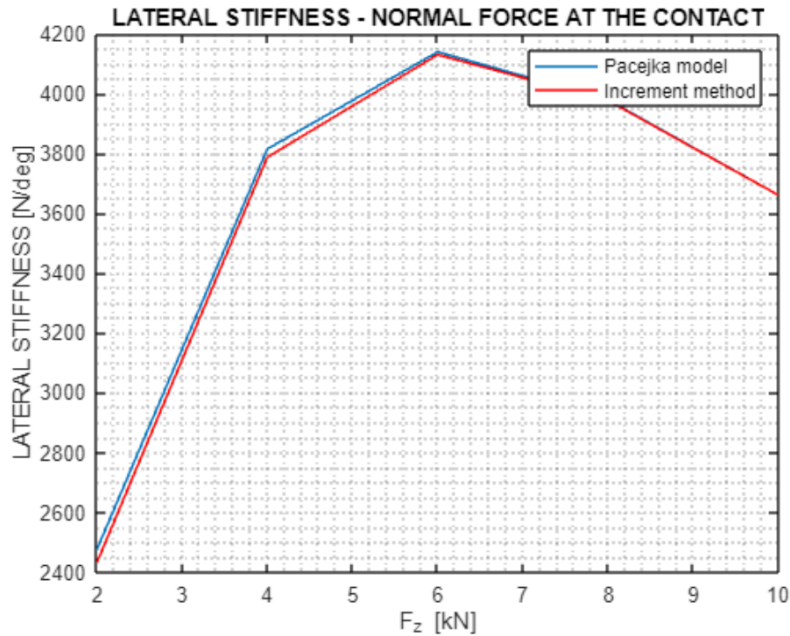
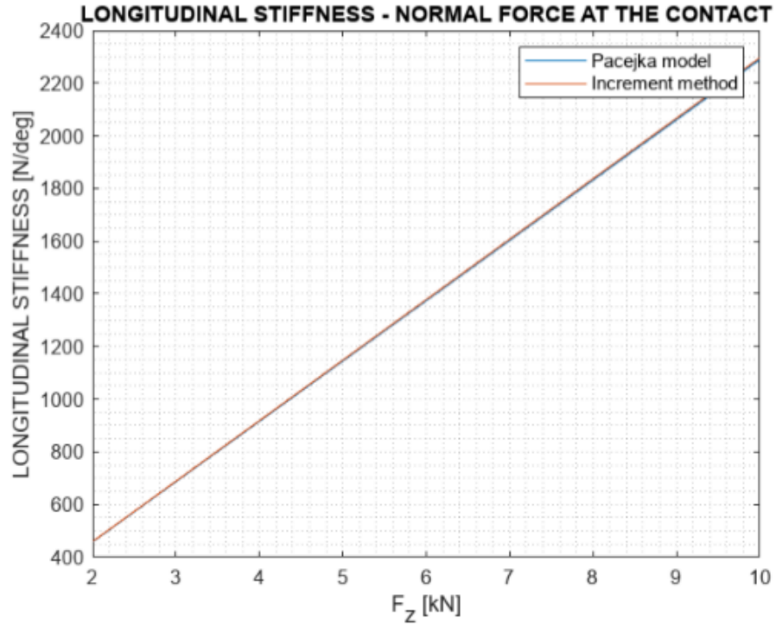
Figure 13: Gough diagram

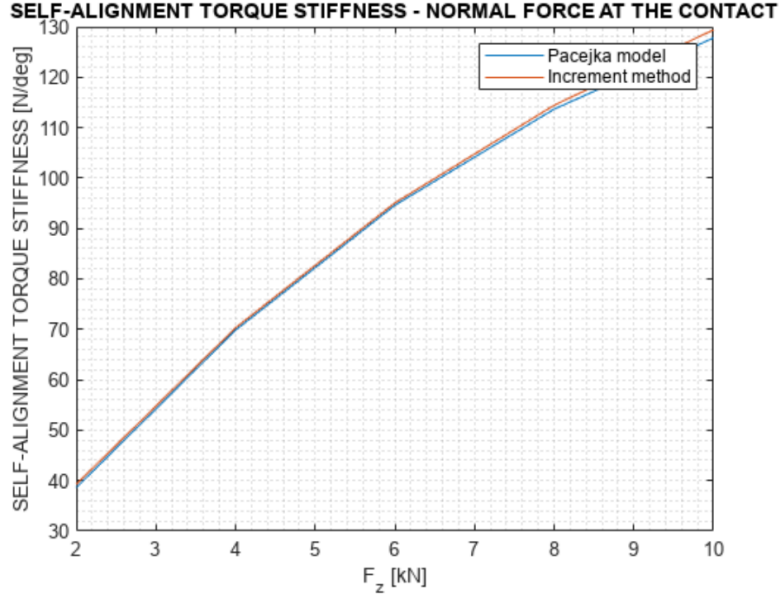
3.5 Tire stiffnesses vs vertical load

This section compares the stiffnesses of the self-aligning torque, side-slip, and longitudinal directions obtained in the first part of the project with those calculated using the incremental method.

The incremental method evaluates a range of values near the origin to compute the stiffnesses as follows:

$$\begin{cases} BCD_x = \Delta F_x / \Delta \sigma \\ BCD_y = \Delta F_y / \Delta \alpha \\ BCD_z = \Delta M_z / \Delta \alpha \end{cases} \quad (10)$$





The values computed using the incremental method are generally similar to those obtained with Pacejka's model, with only a slight deviation observed at higher vertical loads F_z .

4 Interaction between longitudinal and lateral tire-road forces

Pacejka's Magic Formula describes tire behavior assuming that longitudinal and lateral forces act independently. When the tire generates forces along both X' and Y' simultaneously, the behavior changes, as the force along one axis limits the maximum force available along the other axis.

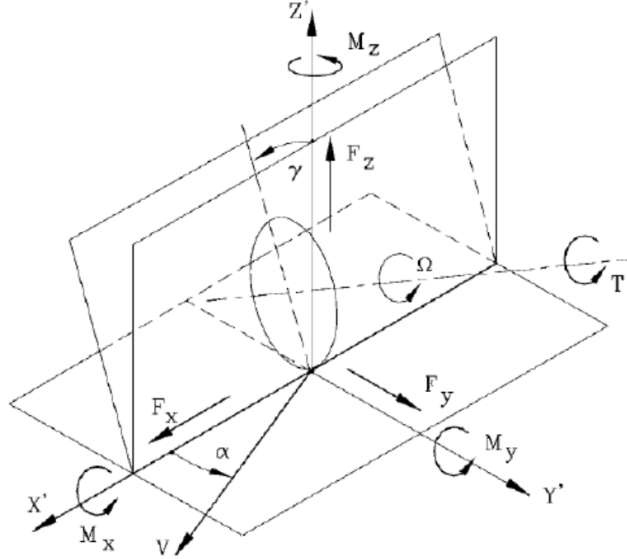


Figure 14: Tire's reference frame

A proper model to approximate the curves $F_y(F_x)$ is the elliptical model:

$$\left(\frac{F_y}{F_{y0}}\right)^2 + \left(\frac{F_x}{F_{x0}}\right)^2 = 1 \quad (11)$$

where F_{y0} is the lateral force F_y at a given sideslip angle α when no longitudinal force F_x is applied, and F_{x0} is the maximum longitudinal force F_x at zero sideslip angle α . The elliptical model gives:

$$F_y = F_{y0} \sqrt{1 - \left(\frac{F_x}{F_{x0}}\right)^2} \quad (12)$$

and consequently:

$$\mu_y = \mu_{y0} \sqrt{1 - \left(\frac{\mu_x}{\mu_{x0}} \right)^2} \quad (13)$$

where μ_y and μ_{y0} represent the lateral friction coefficient with and without any longitudinal force, respectively. The cornering stiffness C is also an important parameter in the tire-road interaction. It is defined as:

$$C = \frac{dF_y}{d\alpha} \quad (14)$$

which is also referred to as the slope BCD_y . The cornering stiffness C depends on the longitudinal force F_x , as can be seen by differentiating the previous equation ($F_y = -C\alpha$).

$$C = C_0 \sqrt{1 - \left(\frac{F_x}{\mu_{xp} F_z} \right)^2} \quad (15)$$

where the coefficient C_0 is the cornering stiffness when no longitudinal force is applied.

Building on the previous discussion of the elliptical tire model and the interaction between longitudinal and lateral forces, the following evaluations and plots are performed:

- the lateral force F_y as a function of the longitudinal force F_x
- the lateral force coefficient μ_y as a function of the longitudinal force coefficient μ_x for different side-slip angles $\alpha = 2^\circ, 4^\circ, 6^\circ, 8^\circ, 10^\circ$, with a constant vertical load $F_z = 4$ kN and zero camber angle $\gamma = 0^\circ$.
- the side-slip stiffness C as a function of the longitudinal force F_x for different vertical loads $F_z = 2, 4, 6, 8, 10$ kN at zero sideslip angle ($\alpha = 0^\circ$)

4.1 Lateral force F_y as a function of the longitudinal force F_x

In a typical tire, both longitudinal and lateral forces act simultaneously, and they can be represented together in a graph with all curves plotted for a constant sideslip angle α . While a polar diagram is usually obtained from experimental tests, in this case it was generated using the elliptical model. This provides a good approximation for values near the Y axis, but deviations occur near the maximum F_x , where the actual behavior differs. For higher values of α (8° and 10°), the curves nearly overlap, indicating that the tire has reached its saturation point.

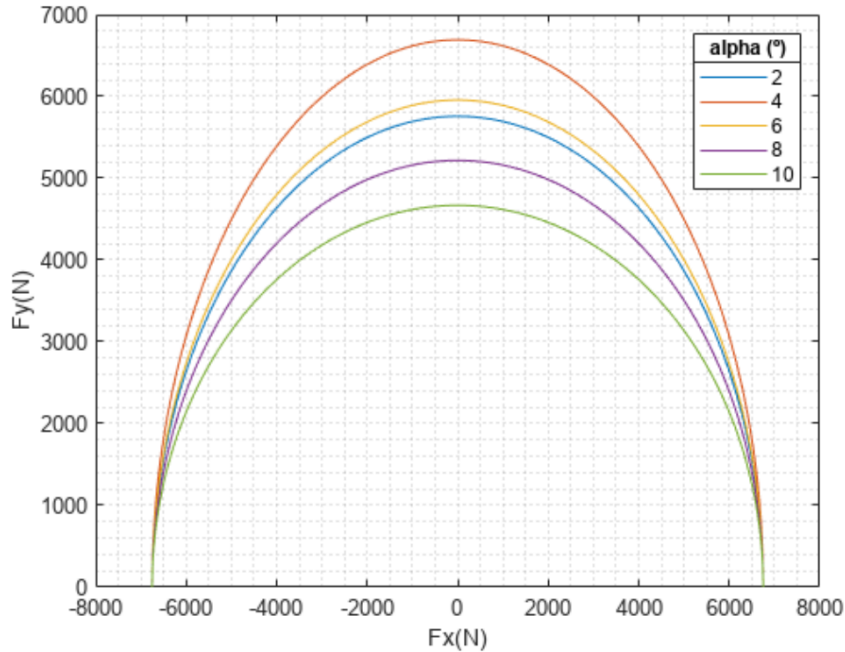


Figure 15: Lateral force vs longitudinal force

4.2 Lateral force coefficient μ_y as a function of the longitudinal force coefficient μ_x

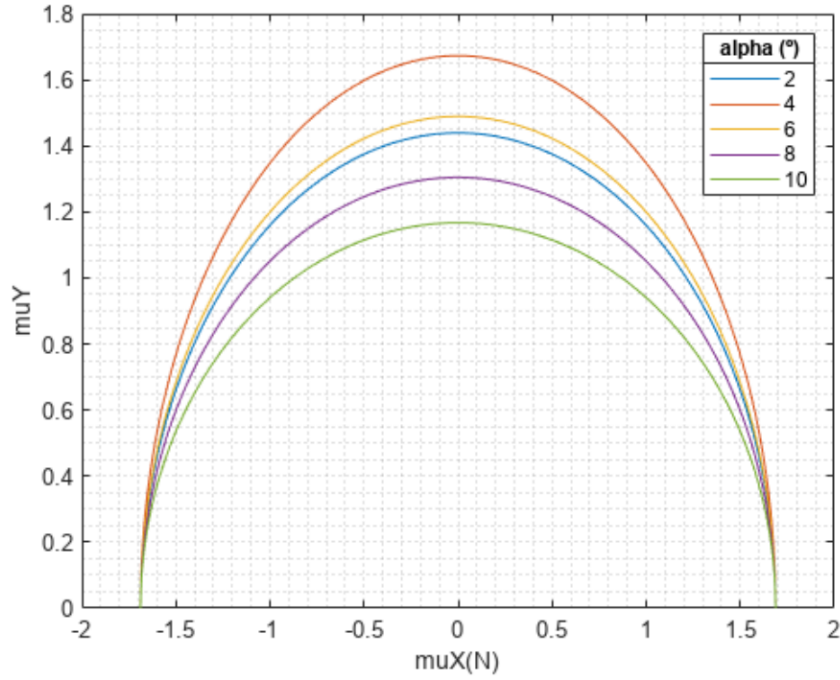


Figure 16: Lateral force coefficient vs longitudinal force coefficient

This graph is similar to the previous one, except that the values are normalized by the vertical load $F_z = 4$ kN to obtain the corresponding friction coefficients.

4.3 Side-slip stiffness C as a function of the longitudinal force F_x

The chart reported in this section is obtained from Equation 15.

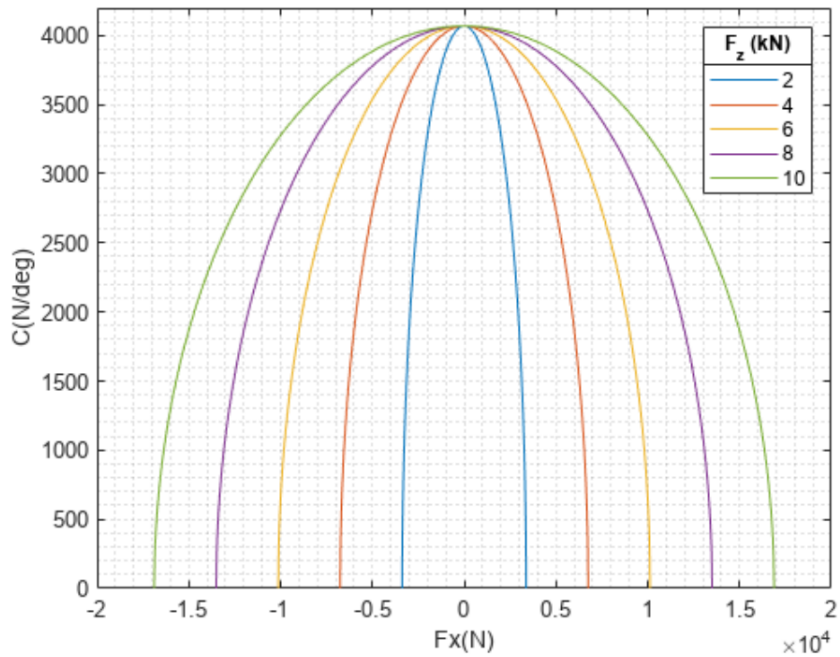


Figure 17: Side-slip stiffness vs longitudinal force

For constant vertical load F_z , an increase in F_x causes the cornering stiffness to decrease. As F_z increases, the tire becomes stiffer, resulting in higher cornering stiffness values, as discussed

in the first part of this project. Moreover, the peak value of the curve is equal to the cornering stiffness when no longitudinal force is applied, i.e. C_0 .

5 Conclusions

This project provided a comprehensive analysis of tire dynamics through both linear and non-linear modeling approaches, using Pacejka's Magic Formula as the foundational tool. The linear model allowed for the evaluation of key stiffness parameters, in particular longitudinal, lateral, and self-aligning torque, under varying vertical loads and camber angles. The non-linear analysis further revealed the behavior of tire forces across a full range of slip ratios and angles, confirming that peak friction coefficients remain largely independent of vertical load, while stiffness values show more sensitivity. The Gough diagram effectively summarized lateral behavior, and the interaction between longitudinal and lateral forces was explored using an elliptical model, highlighting the trade-offs in force generation when both components are active. Overall, the study reinforces the importance of accurate tire modeling in vehicle dynamics and provides valuable insights into how tires respond under diverse operating conditions.